

Metaverse Community Group Loyalty Contagion and Regulation Model Based on User Stickiness

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ABSTRACT

The investigation into the micro-social behaviours of groups in metaverse communities is minimal. This paper proposes a user stickiness-based loyalty contagion and regulation model in a metaverse community inspired by user stickiness in the real world. We hope to establish a link between user stickiness and group loyalty through community regulation, enhance user stickiness, and improve metaverse community loyalty. We verified the feasibility and robustness of the model through simulation experiments and obtained the optimal community regulation intensity through sensitivity analysis experiments. Our results show that communities can maintain healthy long-term development when optimal community regulation intensity is adopted.

Index Terms: Metaverse ecosystem—Social behaviour modelling—User stickiness—Loyalty; Group simulation—Multi-Agents

1 INTRODUCTION

Since the COVID-19 pandemic, people living, studying, and working online have become the norm. The whole of society is migrating to the Internet, accelerating the transformation of society to virtualization [4]. At Present, the metaverse has integrated various new technologies such as Extended Reality, Internet of Things, and Virtual Human, the next generation of Internet applications and virtualization development direction [2, 5]. However, maintaining a healthy and long-lasting development is essential for metaverse communities.

User stickiness is the social behaviour prevalent in natural communities and brands, which directly determines users' loyalty to that community and affects whether the community can develop long-term and healthy [1, 3]. Nevertheless, few scholars have studied the impact of user stickiness on community loyalty in the metaverse ecosystem. To research the inner connection between user stickiness and the existence of metaverse communities, we construct a model for studying the relationship between user stickiness and loyalty. The main contributions of this paper are (1) Different from the macro research perspective of technology and risk. We study the impact of user stickiness on the metaverse community from micro-social behaviour and provide management advice for the healthy development of the metaverse community. (2) The model simulates the behavioural changes of users in the metaverse community. (3) Contrary to the traditional social behaviour modelling approach, the model is based on a multi-agents system and uses modular modelling.

2 METHODOLOGY

We constructed the AIR (Agent Index Regulation) model to make our approach better comprehensible. It contains four modules: environment module, agent module, index module, and regulation

module, as shown in Figure 1. The agent module defines the agent and models the action preferences. The index module calculates the community loyalty index for each moment in the community. The regulation module can maintain the community loyalty index between healthy ranges.

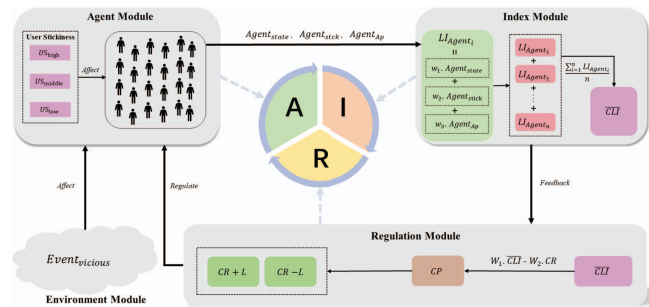


Figure 1: Model frame diagram. The three modules have extremely interrelated and linked relationships with each other, and each module has its internal operating mechanism. They maintain the community loyalty index in a healthy range.

Our study uses the AIR model to simulate user loyalty's impact in metaverse communities due to user stickiness. The community loyalty index is maintained by regulation to prevent the community from falling into chaos and losing users. The process of the model can be defined as follows. (1) The environment module provides activity scenes and influences events for the agents. (2) In the agent module, the agent determines behavioural preferences and changes its area based on user stickiness and the environment. (3) The community loyalty index is calculated automatically, and feedback to the regulation module is in the index module. (4) The regulation module will make regulations based on the community loyalty index to maintain the healthy development of the community.

3 EXPERIMENT

3.1 Simulation Experiment

The primary purposes of the simulation experiments are: (1) We are verifying the feasibility of the model through simulation experiments. (2) We are implementing an animation demonstration of the model through simulation software.

The number of agents can accurately reflect the trend of each type of agent running with the model. We give the simulation experiment results containing the change in the number of agents (Figure 2).

Through the simulation experiment results, we found that when the strength of community regulation (CR) is constant, the model is consistent with the trend of agent change under different samples. When the community regulation is at the default value ($CR = 0.05$), the community is unmanaged at this time. Users will cluster toward people with lower loyalty and gradually lose. We could conclude the following from the simulation experiments. (1) Our model can produce the same experimental results under different samples, and

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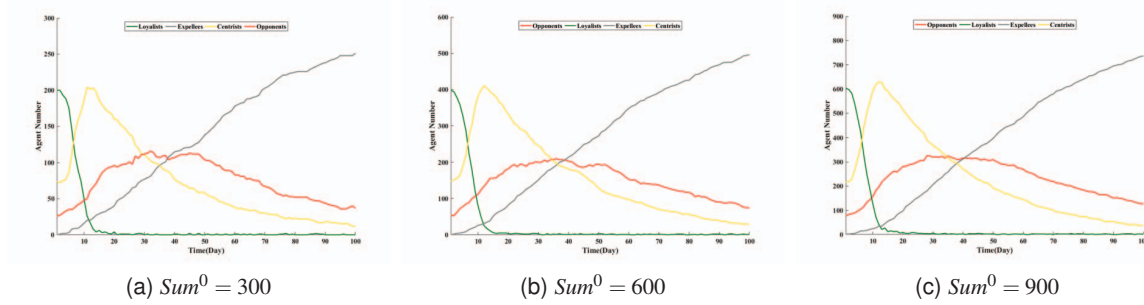


Figure 2: Agent number time graph. In the (a)(b)(c), Sum^0 are different, the community regulation intensity is kept constant($CR = 0.05$).

the model is robust and applicable. (2) The community will be in a highly unhealthy state when unregulated.

3.2 Sensitivity Analysis Experiment

Community regulation is an external force exerted on the group of agents, which can affect the model results individually and is the only parameter used for regulation. The primary purpose of the sensitivity analysis experiment is as follows: (1) Determine the effect of different community regulation efforts on community loyalty. (2) Determine the value of community regulation to optimize the comprehensive performance of the model.

The specific data for the sensitivity analysis experiments are given in Table 1. The experimental data were overly large, and we selected values with particular time nodes(1, 7, 14, 21, 30, 50, 75, 100) in the table.

Table 1: Sensitivity analysis experimental data (community loyalty index)

Days	$CR_{0.05}$	$CR_{0.25}$	$CR_{0.45}$	$CR_{0.65}$	$CR_{0.75}$	$CR_{0.85}$
1	0.5733	0.5783	0.5783	0.5783	0.5783	0.5783
7	0.2304	0.5692	0.5775	0.5894	0.5894	0.5911
14	-0.2585	0.5009	0.5810	0.6053	0.6095	0.6060
21	-0.3659	0.4126	0.5712	0.6288	0.6497	0.6455
30	-0.4388	0.2808	0.5913	0.6399	0.6849	0.7036
50	-0.5872	-0.0716	0.5966	0.7295	0.7857	0.7787
75	-0.6508	-0.2939	0.6230	0.8304	0.8796	0.8720
100	-0.7115	-0.3431	0.6893	0.8800	0.9144	0.9129

Table 1 shows that the community loyalty index did not decline to negative values again when $CR > 0.25$. When $CR \geq 0.45$, the trend of the community loyalty index is no longer evident but still ranges upward. When $CR \geq 0.75$, the community loyalty value finally showed an upward trend, and the community loyalty index was steadily greater than the initial value after 50 days from the end.

Evaluating the effectiveness of regulation should consider user loyalty and the cost of community control. We combine these two metrics in Figure 3 to evaluate the comprehensive performance under different control efforts. We conclude from the sensitivity analysis experiments as follows. (1) Community regulation intensity is not positively correlated with comprehensive performance. (2) The model achieves the best comprehensive performance when the community regulation intensity is 0.75.

4 CONCLUSION AND FUTURE WORK

This paper aims to study social behaviour in the metaverse ecosystem from a microscopic perspective and explore the impact of user stickiness on loyalty in metaverse community clusters. Based on user stickiness, we construct a model of the contagion and regulation

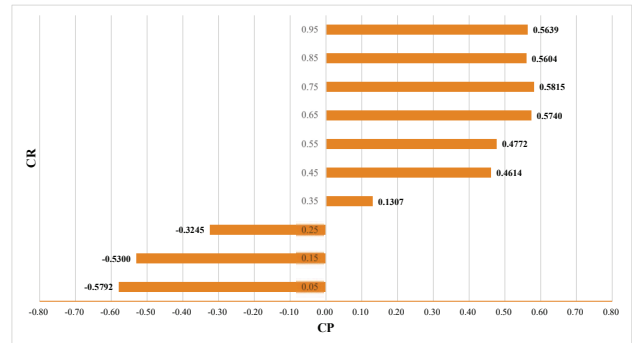


Figure 3: Comprehensive performance diagram. The horizontal coordinates of the graph are the values of the comprehensive performance(CP) and the vertical coordinates are the values of the community regulation(CR). In this case, we use $CP = 0$ as the cut-off point.

of loyalty in the metaverse community. We conduct a sensitivity analysis experiment on the model to obtain the optimal regulation value for community management.

In future work, we will migrate our model to a real-world metaverse community and consider the unique properties of VR, such as perception, immersion, and embodiment.

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